METRIC

MIL-T-50059B
15 August 1983
SUPERSEDING
MIL-T-50059A
21 September 1976

MILITARY SPECIFICATION

TITANOUS SULFATE SOLUTION, 20-PERCENT (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of titanous sulfate solution.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020 - Chemicals, Liquid, Dry, and Paste: Packaging of
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Titanous sulfate content. Titanous sulfate solution shall contain no less than 19 nor more than 22 percent by weight titanous sulfate, $Ti_2(SO_4)_3$, when tested as specified in 4.2.4.1.

3.2 Stabilizer. Titanous sulfate solution shall contain sufficient stabilizer (citric acid or sodium citrate) to prevent hydrolysis of the titanous sulfate when diluted for use. The contractor shall certify that this requirement has been met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the titanous sulfate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For titanous sulfate test. See 6.5 for sampling and testing precautions. Sampling shall be conducted in accordance with table I. A representative specimen of approximately 10 milliliters (mL) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE I. Sampling for titanous sulfate test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects:

AQL 1.0 percent defective

(a) Container damaged or leaking

AQL 2.5 percent defective

(b) Contents per container not as specified
(c) Container not as specified
(d) Container closure not as specified
(e) Marking incorrect, missing, or illegible
(f) Fiberboard packing aids missing or not as specified
(g) Unitization not as specified

4.2.3.2 For titanous sulfate test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.0 percent defective.

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4.2.4 Titanous sulfate tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Titanous sulfate content.

(a) Standard ferric ammonium sulfate solution. Dissolve approximately 50 grams (g) of ferric ammonium sulfate [$FeNH_4(SO_4)_2 \cdot 12H_2O$] in water to make 1 liter of solution. To a 25-mL portion add 2 mL of hydrochloric acid and 2.5 g of potassium iodide and let stand for 30 minutes in the dark. Dilute with 100 mL of water and titrate the liberated iodine with 0.1N sodium thiosulfate solution using starch as the indicator. Calculate the normality of the ferric ammonium sulfate solution as follows:

$$\text{Normality} = \frac{AB}{25}$$

where: A = Milliliters of sodium thiosulfate solution used in the titration
and
B = Normality of the sodium thiosulfate solution.

(b) Procedure. In an inert atmosphere (nitrogen or carbon dioxide) weigh 2 mL of specimen into a small weighing bottle. Add 50 mL of freshly boiled water to a 250-mL Erlenmeyer flask. Allow the water to cool to room temperature and then begin to flush with nitrogen or carbon dioxide. Transfer the weighing bottle and contents to the Erlenmeyer flask. Cautiously add 5 mL of concentrated sulfuric acid. Mix the solution in the flask and then titrate with the standardized ferric ammonium sulfate solution prepared in (a). When the purple color of the solution begins to fade, add 1 mL of 30-percent aqueous ammonium thiocyanate solution and continue the titration until a reddish-brown tint persists for more than 15 seconds. Discontinue the inert gas flush. Calculate the percent by weight titanous sulfate as follows:

$$\text{Percent titanous sulfate} = \frac{19.2\ AB}{W}$$

where: A = Milliliters of ferric ammonium sulfate solution used in the titration,
B = Normality of the ferric ammonium sulfate solution, and
W = Weight of specimen in grams.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes:

(a) Upright
(b) Upside down

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(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Preservation. Titanous sulfate solution shall be preserved level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. A quantity of 1 liter (+10 or -0 mL) of titanous sulfate solution shall be unit packed in accordance with Department of Transportation (DOT) regulations in accordance with the general requirements of PPP-C-2020, in a nominal 1 liter capacity glass bottle cushioned within enclosure A or B. The bottle shall conform to group A, class 1, type d, style 1, grade 1, closure A, with seal A of PPP-C-186. The facing for the liner of the screw-cap shall be formed from phenol-formaldehyde resin or an alternate material proven to be at least as suitable for use in packaging titanous sulfate solution. The screw-cap shall be closed to a torque within the range as specified by the bottle manufacturer. There shall be no evidence of leakage when tested as specified in 4.2.5.

5.1.1.2 Intermediate packing. Twelve bottles of titanous sulfate solution unit packed as specified in 5.1.1.1 shall be intermediately packed with closures uppermost in a close-fitting fiberboard box conforming to type CF, class domestic, variety DW, grade 200 of PPP-B-636. The box shall be furnished with close-fitting, full-bottle-height partitions and with liners, top pads, and bottom pads conforming to type CF, class domestic, variety DW, grade 200 of PPP-F-320. Motion of the contents within the box shall be prevented by insertion of additional pads, as needed. The box shall be closed in accordance with method I or II of PPP-B-636.

5.1.2 Level C. A quantity of 1 liter (+10 or -0 mL) of titanous sulfate solution shall be unit packed level C in a glass bottle in a manner to assure maintenance of the specified purity and quantity of contents for a minimum duration of 6 months, compliance with DOT regulations, and compliance with the general level C requirements of PPP-C-2020.

5.2 Packing. Titanous sulfate solution shall be packed level A, B, or C as specified (see 6.2).

5.2.1 Level A. One intermediate pack prepared as specified in 5.1.1.2 shall be packed with unit pack closures uppermost in a close-fitting box. The box shall conform to class 3, style optional of PPP-B-585; type overseas, style A, B, or I, grade A of PPP-B-601; or class 2, style 4, grade A of PPP-B-621. The

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box shall be for a type I load and shall be closed and reinforced as specified in the applicable box specification. Motion of contents within the box shall be prevented by insertion of fiberboard pads conforming to grade V3c of PPP-F-320, as needed.

5.2.2 Level B. Titanous sulfate shall be packed level B in the same manner as specified for level A, except that the box shall conform to class weather-resistant, grade V3c of PPP-B-636.

5.2.3 Level C. Containers of titanous sulfate solution shall be packed level C in accordance with DOT regulations, to assure acceptance by common carrier, and in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules and to assure safe delivery from supply source to the first destination.

5.2.4 Unitized loads. Uniform quantities of level A or B packs of titanous sulfate shall be palletized with unit pack closures uppermost in accordance with the applicable requirements of load type I of MIL-STD-147, using the soft-wood pallet conforming to type IV of NN-P-71. Level C packs shall be unitized in a manner to assure acceptance by common carrier and safe delivery to destination.

5.3 Marking. Level A and C unit and intermediate packs; level A, B and C packs; and unitized loads shall be marked in accordance with MIL-STD-129. Each container shall be marked to show lot or batch number, the date of manufacture of the titanous sulfate, and the following precautionary information:

DANGER - CAUSES SEVERE BURNS
Do not get liquid or vapor in eyes, on skin or clothing. In case of contact, immediately flush skin or eyes with large amounts of water. Obtain medical attention. Use only as dye stripping agent in laundry operations in compliance with activity directives. Protective gloves, goggles and aprons shall be worn by all personnel handling this agent in laundry operations.

SHADE FROM RADIANT HEAT
KEEP DRY

6. NOTES

6.1 Intended use. Titanous sulfate is intended for use as a stripping agent in laundries.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and the level of preservation and packing required (see 5.1 and 5.2).

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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD
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Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-50059B	TITANOUS SULFATE SOLUTION, 20-PERCENT (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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